

User Guide

Project: Enterprise Quantum Computing Platform — QFT Bank **Student:** Syed Muhammad Tahoor Ali **Diploma:** EduQual Level 6, Diploma in Artificial Intelligence Operations (ANPP-OP)

1. Getting Started

Open the platform in a browser at the deployment URL (in the Kubernetes deployment, `http://<minikube-ip>:30080`). The application is a single-page web app with a sidebar listing the modules. A status indicator confirms the backend is reachable. Select a module from the sidebar to open its screen; each screen is self-contained and can be used independently.

This guide walks through each module as a user of the web interface. No command line is needed.

2. Quantum Circuit Designer

Builds and runs a quantum circuit and shows the measurement results.

How to use it. Choose the number of qubits. Add gates one at a time from the supported set (Hadamard, Pauli X/Y/Z, S, T, the rotation gates rx/ry/rz, and the multi-qubit CX, CZ, SWAP, and Toffoli). For each gate, specify the qubit indices it acts on — the input accepts a list such as `0, 1`. Rotation gates additionally take an angle. Set the number of shots (measurement repetitions), then run.

What you see. The Results panel shows a text rendering of the assembled circuit, a histogram of measured bitstrings, and the probability distribution. A classic first experiment is a Bell state: apply Hadamard to qubit 0, then CX from qubit 0 to qubit 1, and observe an approximately even split between `00` and `11` — the signature of entanglement.

3. Quantum Algorithms

Demonstrates four enterprise-relevant quantum algorithms, each with its own controls.

Grover's search. Enter the number of qubits and the target bitstring (for example `101`). Grover amplifies the marked state so it dominates the measured distribution, finding it in far fewer evaluations than checking each possibility in turn.

Quantum Fourier Transform. Enter a number of qubits and an input basis state (for example `110`). The QFT is the quantum analogue of the discrete Fourier transform and underlies several important algorithms.

QAOA (Max-Cut). Provide a graph as a set of edges and a node count, with optional layer count and iteration cap. QAOA searches for a good cut of the graph; the result shows the best partition found.

VQE. Enter a qubit count and iteration cap. The Variational Quantum Eigensolver estimates a lowest-energy (ground-state) value by iteratively adjusting circuit parameters.

What you see. Each run shows the algorithm's output and execution statistics. Because these are genuine simulations, results reflect real quantum behaviour rather than pre-set answers.

4. Financial Portfolio Optimization

Optimizes a simulated investment portfolio and benchmarks the quantum result against an exact classical answer.

How to use it. Set the number of assets (up to eight), the budget (exactly how many assets to select), and optionally the risk-aversion and other tuning parameters. A seed keeps the simulated market reproducible. Run the optimization.

What you see. Three panels. *Simulated Market Data* shows the generated assets with their returns and volatilities. *Classical (Brute Force)* shows the exact optimum found by checking every combination — feasible because the asset count is capped. *Quantum (QAOA)* shows the solution the quantum optimiser reached, and whether it matched the classical optimum. This side-by-side is deliberately honest: it lets you verify the quantum result against ground truth rather than asserting an unfalsifiable advantage.

5. Executive Operations Dashboard

Aggregates activity from the other modules for a leadership view. It runs no new quantum computation.

What you see. Success rates by module, experiment counts by module, and a recent-activity feed drawn from the shared execution history. This is the screen intended for non-technical stakeholders who want the state of the programme at a glance rather than the details of any single run.

6. Post-Quantum Cryptography Readiness

An educational and governance screen on the quantum threat to cryptography and the path to quantum-safe algorithms.

What you see. A reference of current public-key algorithms and their vulnerability to quantum attack; the NIST post-quantum standards (ML-KEM, ML-DSA, SLH-DSA, HQC); a conceptual demonstration of quantum phase estimation (the mechanism behind Shor's algorithm, which threatens current cryptography); a simulated inventory scan that assigns risk scores to systems; an organizational risk assessment; and a phased migration plan. Together these show why an organisation should begin planning for post-quantum cryptography before large-scale quantum computers arrive.

7. Multi-Framework Comparison

Demonstrates that results are correct across quantum frameworks, not an artefact of one library.

What you see. The same Bell state constructed and run in Qiskit, PennyLane, and Cirq, with the three results shown side by side. Agreement across all three confirms framework-independent correctness and satisfies the specification's requirement to demonstrate PennyLane and Cirq alongside Qiskit.

8. Classical ML vs Quantum Optimization

Compares a classical machine-learning approach to asset selection against the quantum approach.

How to use it. Set the asset count, budget, and the number of simulated market scenarios used to train the model. Run the comparison.

What you see. The model training summary, the classical ML prediction on a fresh evaluation market, and the exact optimum for comparison. This situates the quantum approach against a classical learned baseline as well as against brute force.

9. Market Analytics

Presents a simulated market through three complementary visualisations.

How to use it. Set the number of assets and a seed, then generate.

What you see. A Pandas-derived summary table (per-asset return, volatility, correlation, return/risk ratio); an interactive chart (hover for exact figures, drag to zoom, double-click to reset); and a static chart image suitable for embedding in a report. The three views answer different questions — the numbers, interactive exploration, and a fixed printable snapshot respectively.

10. Tips

- The seed fields make runs reproducible: the same seed always produces the same simulated market, which is useful when comparing runs or preparing a demonstration.
- Qubit and asset counts are capped to keep simulations fast on modest hardware and to keep the classical baselines exactly computable.
- Every run that produces history is recorded and will appear in the Executive Dashboard's aggregated view.